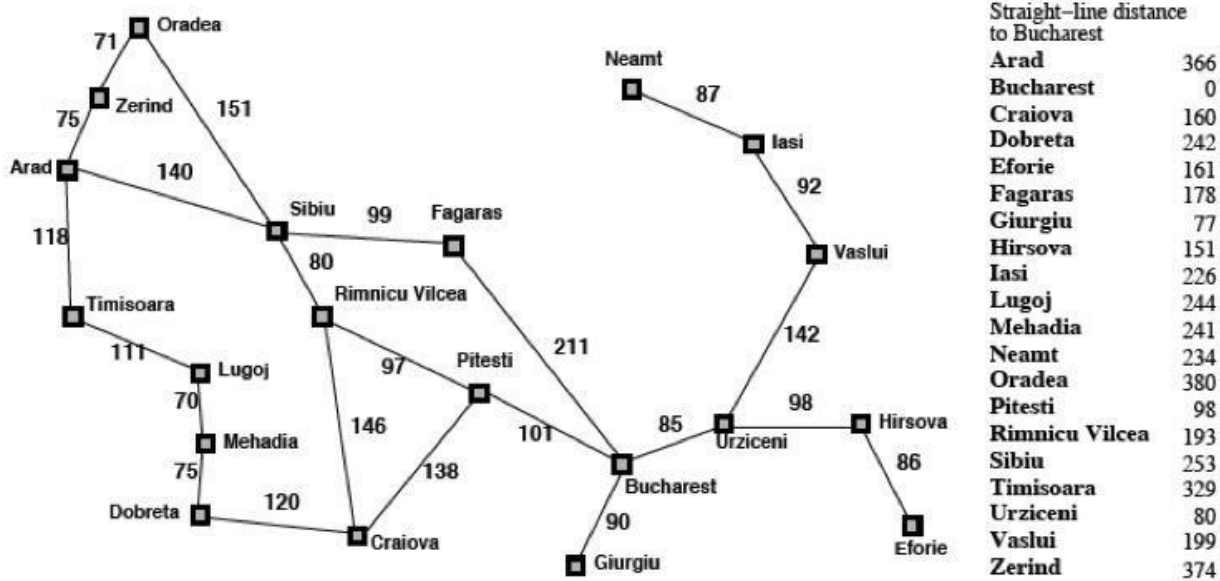


Romania with step costs in km



PROBLEM SCENARIO

On holiday, a flight currently wants to travel to Bucharest from Arad. But there is no direct way to Bucharest from Arad. However, the cities are connected with each other like a graph. The distance between the connected cities are given. The flight wants to travel through the most optimal way. To find the optimal path to travel, another information is provided: the straight line distance between any city and the final destination (Bucharest).

Now help the flight to determine the most optimal value with A* search for the route Arad to Bucharest. You have to use the straight line distance as the heuristic value for the cities.

City	Heuristic value	City	Heuristic value
------	-----------------	------	-----------------

For simplicity assume these notations

Arad	A	Neamt	F
Bucharest	Z	Oradea	B
Craiova	S	Pitesti	P
Eforie	T	Rimnicu Vilcea	R
Fagaras	O	Timisoara	C
Dobreta	V	Urziceni	D
Hirsova	N	Vaslui	H
Iasi	Q	Zerind	E
Lugoj	G		
Mehadia	L		

INPUTS

Your txt file should take each node followed by each destination it can reach and their corresponding distance and heuristics. You are to read the file then ask the user to input the starting and the destination point.

OUTPUTS

The output will contain the total distance from the starting point to the destination followed by printing the nodes it followed to calculate the distance.

SAMPLE INPUT

In the text file:

Arad 366 Zerind 75 Sibiu 140 Timisoara 118

Zerind 374 Arad 75 Oradea 71

Oradea 380 Zerind 71 Sibiu 151

... ..

Bucharest 0 Pitesti 101 Fagaras 211 Giurgiu 90 Urziceni 85

Start node: Arad
Destination: Bucharest

Sample output

Path: Arad -> Sibiu -> Rimnicu -> Pitesti -> Bucharest
Total distance: 418 km

If there is no path found from the Start node to the End node, simply print "NO PATH FOUND"